

1. What is branching in Git and how do you merge a branch into the main branch?

Answer : A branch is a separate copy of the code where we can work on new features or fixes without affecting the main project. After finishing the work, switch to the main branch and merge it using:

```
git checkout main  
git merge branch-name
```

2. What is a merge conflict in Git and how can it be resolved?

Answer : A merge conflict happens when the same part of a file is changed in different branches. Git cannot decide which change to keep. We resolve it by editing the file, removing the conflict markers, saving the correct code, then running:

```
git add .  
git commit
```

3. How do you connect and push code to a remote Git repository?

Answer : First, connect the local repository to the remote repository:

```
git remote add origin <repository-url>
```

Then push the code using:

```
git push -u origin main
```

4. What are GitHub, GitLab, and Bitbucket?

Answer : GitHub, GitLab, and Bitbucket are Git hosting platforms. They let us store code online, work with teams, track changes, review code, and manage projects using Git.

5. What are Git hooks, submodules, and subtrees?

- Git Hooks: Scripts that run automatically before or after Git actions like commit or push.
- Submodules: Allow one Git repository to be included inside another while keeping its own history.
- Subtrees: Copy another repository into the current project and manage it without using a separate repository link.

6. What are the benefits of contributing to open-source projects and how can you contribute?

Answer : Contributing to open-source helps improve coding skills, gain real project experience, learn from other developers, and build a strong portfolio. We can contribute by fixing bugs, adding new features, improving documentation, testing projects, or reviewing pull requests.